Project

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Background

Problem Formulation

The NBA Draft is an annual event where all 30 NBA teams select eligible players to join their rosters, serving as the primary pathway for new talent to enter the league. The draft consists of two rounds, with each team holding one pick per round, for a total of 60 selections. The order in which teams pick is not arbitrary: teams that performed worse in the previous season are rewarded with earlier picks, giving struggling franchises the opportunity to rebuild through high-potential young talent. Specifically, the 14 lowest-ranked teams enter a weighted lottery to determine the first 14 picks of the first round, with worse records receiving higher odds. The remaining picks in both rounds follow reverse standings order — the worst team picks first, the best team picks last. Prior to the draft, teams evaluate prospects through the NBA Draft Combine, private workouts, and scouting reports. Scouts assess a wide range of factors including physical measurements, athletic testing, skill level, and projected development trajectory. The result is an internal team ranking that ultimately determines which player a team selects when their pick arrives.

Pick position carries substantial consequences for a player's career. First-round selections receive fully guaranteed multi-year contracts, whereas second-round picks are not guaranteed and must compete for roster spots with little job security. This distinction makes the first-round versus second-round boundary a meaningful and practically important threshold to study.

Given this context, understanding what physical attributes are associated with first-round selection is of both scientific and practical interest. Height, in particular, has long been regarded as a fundamental advantage in basketball — taller players are better positioned to score near the basket, contest shots, and control the boards. It is therefore natural to ask whether teams systematically favour taller players when making first-round selections. Our scientific question is:

> **Does player height have a positive effect on the probability of being selected in the first round of the NBA Draft?**

Mathematically, we model this via Bayesian logistic regression, where the binary outcome (first round = 1, second round = 0) is regressed on player height. The key quantity of interest is the **posterior probability that the slope on height is positive**:

$$P(\beta_{\text{height}} > 0 \mid \mathbf{Y})$$

Team Contributions

Public repository link: https://github.com/ymyms/stat-405-project

Literature Review

A growing body of literature examines how NBA teams evaluate talent and make draft decisions, focusing on player performance and observable characteristics. A closely related study is Greer et al. (2019), which uses a logistic regression framework to analyze the determinants of NBA draft selection. The authors find that scoring ability is the most important predictor of whether a player is selected, while other factors such as team success and player characteristics also play a role. Complementing this, Cui et al. (2019) analyze NBA Draft Combine data and show that anthropometric characteristics, particularly height and wingspan, are key factors distinguishing drafted from undrafted players. Their findings highlight the importance of physical attributes in talent identification and suggest that taller players have a systematic advantage in draft outcomes.

Despite these contributions, existing studies primarily focus on whether a player is drafted at all, rather than distinguishing between first-round and second-round selection, which represents a meaningful economic threshold due to differences in contract guarantees. Furthermore, most prior research relies on frequentist methods and emphasizes statistical significance rather than probabilistic interpretation. As a result, these approaches do not directly quantify the probability that a specific attribute, such as height, has a positive effect on draft outcomes.

This paper contributes to the literature by focusing on the probability of being selected in the first round and adopting a Bayesian logistic regression framework. This approach allows for direct estimation of the posterior probability that height has a positive effect on draft selection, providing a more intuitive and decision-relevant interpretation. Using a publicly available NBA player dataset from Kaggle, this study offers a complementary perspective on how physical attributes influence draft outcomes.

Data Cleaning

```
nba <- read.csv("all_seasons.csv")
```

The raw dataset contains 12,844 player-season observations across 22 variables, spanning the 1996–97 to 2022–23 NBA seasons. Each row represents one player in one season, meaning the same player may appear multiple times across different years.

```
recent_20 <- nba[nba$draft_year>=2006 & !(nba$draft_round== "Undrafted") & !(nba$draft_number== "Undraf
rookie_data <- recent_20 |>
  dplyr::arrange(player_name, season) |>
  dplyr::distinct(player_name, .keep_all = TRUE)
rookie_data$draft_round <- ifelse(rookie_data$draft_round==1,1,0)
```

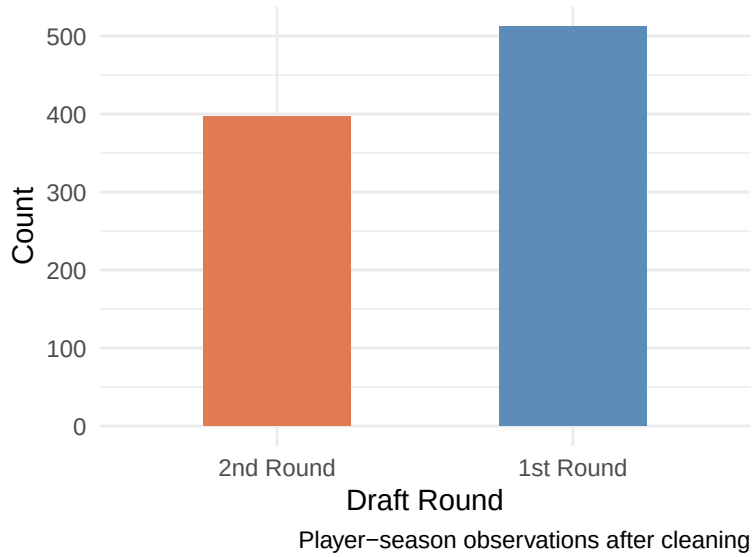
Four cleaning steps were applied to prepare the data for modelling. First, we restricted the data to players drafted in 2006 or later, retaining only the past decade of draft classes. This ensures that our findings reflect the preferences and evaluation criteria of the modern NBA. Second, players recorded as undrafted were excluded, since our research question concerns the distinction between first- and second-round picks specifically, undrafted players represent a qualitatively different pathway into the league. Third, to avoid the statistical issue of repeated observations per player (the same player's height and draft round appearing across multiple seasons), we retained only each player's earliest season. This gives one observation per unique player, ensuring the independence assumption of logistic regression is satisfied. Fourth, the `draft_round` variable was recoded as a binary indicator: 1 for first-round selections and 0 for second-round selections.

EDA

The goal of this section is to examine the distribution of player height and its relationship with draft round.

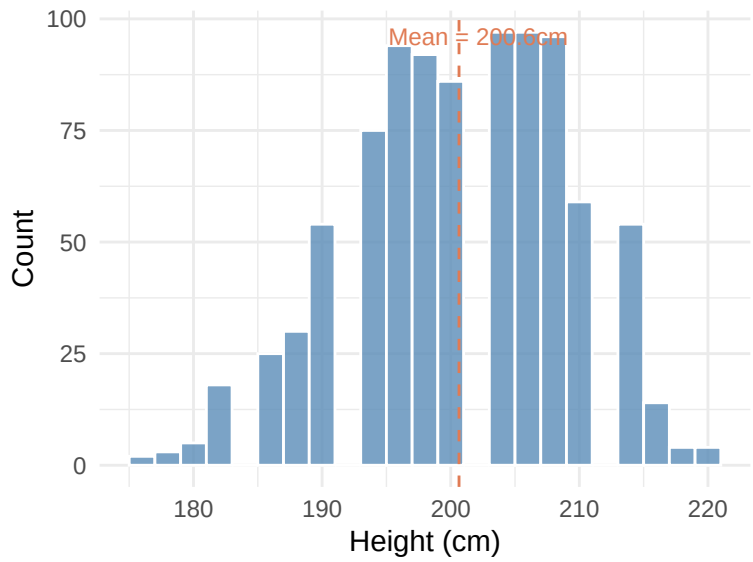
Distribution of Draft Round

```
ggplot(rookie_data,
       aes(x = factor(draft_round, labels = c("2nd Round", "1st Round")))) + geom_bar(fill = c("#E07B54
  labs(x = "Draft Round", y = "Count",
       caption = "Player-season observations after cleaning")+
  theme_minimal()
```



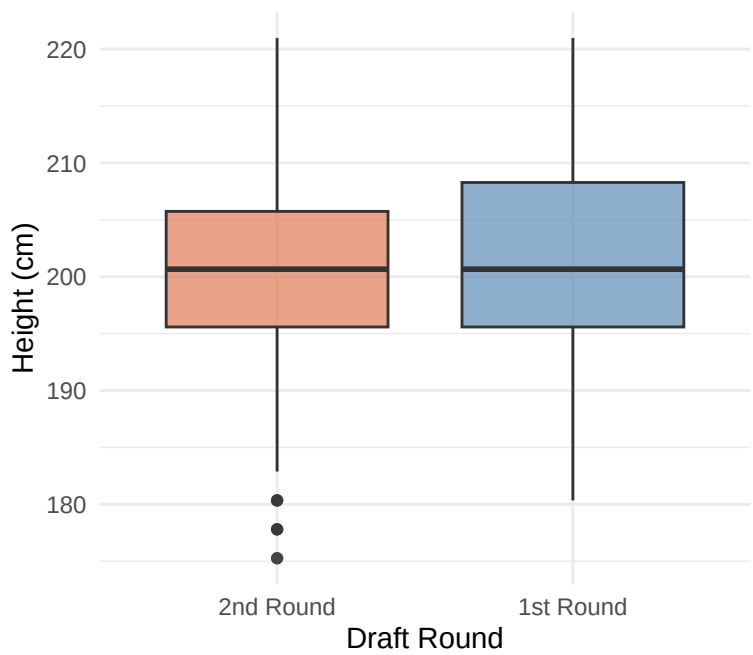
Distribution of Player Height

```
ggplot(rookie_data, aes(x = player_height)) +
  geom_histogram(binwidth = 2, fill = "#5B8DB8", color = "white", alpha = 0.8) +
  geom_vline(xintercept = mean(rookie_data$player_height),
             linetype = "dashed", color = "#E07B54") +
  annotate("text", x = mean(rookie_data$player_height) + 1.5,
           y = Inf, vjust = 2,
           label = paste0("Mean = ",
             round(mean(rookie_data$player_height), 1), "cm"),
           color = "#E07B54", size = 3) +
  labs(x = "Height (cm)", y = "Count") + theme_minimal()
```



Height by Draft Round

```
ggplot(rookie_data,
       aes(x = factor(draft_round, labels = c("2nd Round", "1st Round")),
           y = player_height,
           fill = factor(draft_round))) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  scale_fill_manual(values = c("#E07B54", "#5B8DB8")) +
  labs(x = "Draft Round", y = "Height (cm)") + theme_minimal()
```



```r
rookie_data %>%
  group_by(draft_round) %>%
  summarise(
    n        = dplyr::n(),
    mean_ht  = round(mean(player_height), 2),
    sd_ht    = round(sd(player_height), 2),
    med_ht   = round(median(player_height), 2)
  ) %>%
  mutate(draft_round = ifelse(draft_round == 1,
                              "1st Round", "2nd Round")) %>%
  rename(`Draft Round` = draft_round, N = n,
         Mean = mean_ht, SD = sd_ht, Median = med_ht)
```

```
## # A tibble: 2 x 5
##   `Draft Round`     N  Mean    SD Median
##   <chr>         <int> <dbl> <dbl>  <dbl>
## 1 2nd Round       397  200.  8.63   201.
## 2 1st Round       512  201.  8.16   201.
```

First-round picks are on average taller than second-round picks, though the difference is modest and the distributions overlap considerably. The EDA is consistent with a positive but uncertain slope in the logistic regression model.

```r
library(leaps)
temporary_data <- rookie_data[,-c(1,2,3,7,8,22)]

temporary_data[] <- lapply(temporary_data, as.numeric)
forward <- regsubsets(draft_round ~ .,
                        data = temporary_data ,
                        nvmax = 16,
                        method = "forward")

ss = summary(forward)
ss$which
```

```
##    (Intercept)   age player_height player_weight draft_year draft_number    gp
## 1         TRUE FALSE         FALSE         FALSE      FALSE         TRUE FALSE
## 2         TRUE  TRUE         FALSE         FALSE      FALSE         TRUE FALSE
## 3         TRUE  TRUE         FALSE         FALSE      FALSE         TRUE FALSE
## 4         TRUE  TRUE         FALSE          TRUE      FALSE         TRUE FALSE
## 5         TRUE  TRUE         FALSE          TRUE      FALSE         TRUE FALSE
## 6         TRUE  TRUE         FALSE          TRUE      FALSE         TRUE FALSE
## 7         TRUE  TRUE          TRUE          TRUE      FALSE         TRUE FALSE
## 8         TRUE  TRUE          TRUE          TRUE      FALSE         TRUE FALSE
## 9         TRUE  TRUE          TRUE          TRUE      FALSE         TRUE FALSE
## 10        TRUE  TRUE          TRUE          TRUE      FALSE         TRUE FALSE
## 11        TRUE  TRUE          TRUE          TRUE      FALSE         TRUE FALSE
## 12        TRUE  TRUE          TRUE          TRUE      FALSE         TRUE  TRUE
## 13        TRUE  TRUE          TRUE          TRUE      FALSE         TRUE  TRUE
## 14        TRUE  TRUE          TRUE          TRUE       TRUE         TRUE  TRUE
## 15        TRUE  TRUE          TRUE          TRUE       TRUE         TRUE  TRUE
##      pts   reb   ast net_rating oreb_pct dreb_pct usg_pct ts_pct ast_pct
```

```
## 1  FALSE FALSE FALSE      FALSE    FALSE    FALSE   FALSE  FALSE   FALSE
## 2  FALSE FALSE FALSE      FALSE    FALSE    FALSE   FALSE  FALSE   FALSE
## 3   TRUE FALSE FALSE      FALSE    FALSE    FALSE   FALSE  FALSE   FALSE
## 4   TRUE FALSE FALSE      FALSE    FALSE    FALSE   FALSE  FALSE   FALSE
## 5   TRUE FALSE  TRUE      FALSE    FALSE    FALSE   FALSE  FALSE   FALSE
## 6   TRUE FALSE  TRUE      FALSE     TRUE    FALSE   FALSE  FALSE   FALSE
## 7   TRUE FALSE  TRUE      FALSE     TRUE    FALSE   FALSE  FALSE   FALSE
## 8   TRUE FALSE  TRUE      FALSE     TRUE    FALSE   FALSE   TRUE   FALSE
## 9   TRUE FALSE  TRUE      FALSE     TRUE     TRUE   FALSE   TRUE   FALSE
## 10  TRUE FALSE  TRUE      FALSE     TRUE     TRUE   FALSE   TRUE    TRUE
## 11  TRUE FALSE  TRUE      FALSE     TRUE     TRUE    TRUE   TRUE    TRUE
## 12  TRUE FALSE  TRUE      FALSE     TRUE     TRUE    TRUE   TRUE    TRUE
## 13  TRUE FALSE  TRUE       TRUE     TRUE     TRUE    TRUE   TRUE    TRUE
## 14  TRUE FALSE  TRUE       TRUE     TRUE     TRUE    TRUE   TRUE    TRUE
## 15  TRUE  TRUE  TRUE       TRUE     TRUE     TRUE    TRUE   TRUE    TRUE
```

```r
which.max(ss$adjr2)
```

```
## [1] 7
```

The preditors that are chosen using forward selection and maximum R squared are age,player_height,player_weight,draft_nun pts, ast, and oreb_pct.

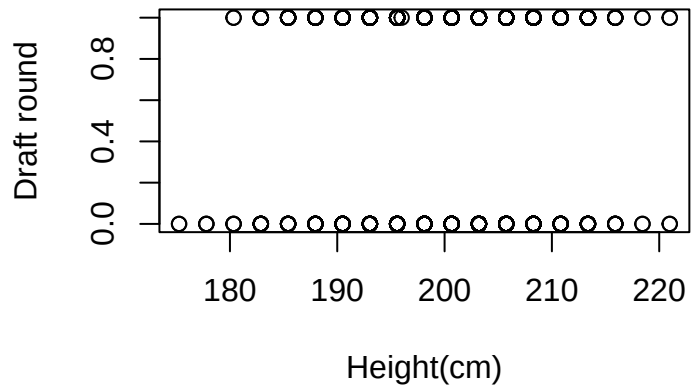
Introduce too many linear dependencies and would introduce far more than 19 predictors after dummy encoding.

```r
glm(formula = draft_round~ age+player_height+player_weight+draft_number+pts+ast+oreb_pct,
       data = temporary_data, family = binomial)%>%  tidy(exponentiate = F)
```

```
## # A tibble: 8 x 5
##   term          estimate std.error statistic  p.value
##   <chr>            <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    38.7       8.71       4.45  8.76e- 6
## 2 age            -0.709     0.154     -4.61  4.12e- 6
## 3 player_height  -0.0245    0.0399    -0.614 5.39e- 1
## 4 player_weight  -0.0169    0.0276    -0.612 5.40e- 1
## 5 draft_number   -0.574     0.0609    -9.41  4.75e-21
## 6 pts             0.138     0.101      1.37  1.70e- 1
## 7 ast            -0.342     0.388     -0.882 3.78e- 1
## 8 oreb_pct        9.76      5.30       1.84  6.57e- 2
```

In the frequentist interpretation, since the p value for slope is smaller than 0.05 (0.029), and the slope is larger than 0 (0.027), we could conclude that there is a statistically significant positive association between the player's height and draft round. We could like to also verify this idea using the Bayesian approach.

```r
plot(y=rookie_data$draft_round,x=rookie_data$player_height, ylab="Draft round",xlab="Height(cm)")
```



From the scatter plot, we can see that when player's height is below 185cm, it is very unlikely for the player to be drafted in the first round. Howeverm after 185cm, the nuisance becomes less obvious.

The model has distribution as following.

$$\text{slope} \sim N(0, 10)\text{intercept} \sim N(0, 10)\theta(\text{height}) = \text{logistic}(\text{intercept} + \text{height} \cdot \text{slope})Y \sim \text{Bernoulli}(\theta(\text{height}))$$

Model Fitting

```
library(distr)

posterior = function(ppl_function, number_of_iterations) {
  log_numerator = -Inf
  log_denominator = -Inf

  for (i in 1:number_of_iterations) {
    log_weight <<- 0.0
    g_i = ppl_function()
    log_numerator = logsumexp(log_numerator, log_weight + log(g_i))
    log_denominator = logsumexp(log_denominator , log_weight)
  }
  return(exp(log_numerator-log_denominator))
}

logsumexp <- function(a,b){
  return(log(exp(a)+exp(b)))
}

###### contents of the scaffold

suppressPackageStartupMessages(library(distr))

## Utilities to make the distr library a bit nicer to use
```

```r
p <- function(distribution, realization) {
  d(distribution)(realization) # return the PMF or density
}

Bern <- function(p) {
  function(y_vec) {
    # y_vec: vector of 0/1 observations
    prod(p^y_vec * (1 - p)^(1 - y_vec))
  }
}

## Key functions called by simPPLe programs

# Use simulate(distribution) for unobserved random variables
simulate <- function(distribution) {
  r(distribution)(1) # sample once from the given distribution
}

# Use observe(realization, distribution) for observed random variables
observe = function(realization, distribution) {
  # `<<-` lets us modify variables that live in the global scope from inside a function
  weight <<- weight * p(distribution, realization)
}

observe_all <- function(y_vector, p_vector) {
  p_vec <- pmax(pmin(p_vector, 1 - 1e-20), 1e-20)
  return((sum(dbinom(y_vector,1,p_vector,log=TRUE))))
}
```

```r
set.seed(405)

hierarchical <- function() {
  center <- simulate(Norm(0,1))
  spread  <- simulate(Exp(1))
  intercept <- simulate(Norm(0,1))
  slope_height <- simulate(Norm(center,spread))
  slope_weight <- simulate(Norm(center,spread))
  slope_pts <- simulate(Norm(center,spread))
  slope_ast <- simulate(Norm(center,spread))
  slope_draft <- simulate(Norm(center,spread))
  slope_age <- simulate(Norm(center,spread))
  slope_oreb <- simulate(Norm(center,spread))

  log_likelihood <- observe_all(temporary_data$draft_round,
              plogis(intercept +
                 slope_height * temporary_data$player_height+
                 slope_draft * temporary_data$draft_number +
                 slope_age * temporary_data$age+
                 slope_weight * temporary_data$player_weight +
                 slope_pts * temporary_data$pts +
                 slope_ast * temporary_data$ast +
                 slope_oreb * temporary_data$oreb_pct))
```

```r
  log_weight <<- log_weight + log_likelihood

  return(ifelse(slope_height > 0, 1, 0))
}

posterior(hierarchical,100)
```

```
## [1] NaN
```

```r
data_complicated <- list(
  N = nrow(temporary_data),
  height = temporary_data$player_height,
  y = rookie_data$draft_round,
  draft_number = temporary_data$draft_number,
  age = temporary_data$age,
  player_weight = temporary_data$player_weight,
  pts = temporary_data$pts,
  ast = temporary_data$ast,
  oreb_pct = temporary_data$oreb_pct)


mode <- cmdstan_model("complicated_bernoulli.stan")
```

```
## Warning in readLines(stan_file): incomplete final line found on
## 'complicated_bernoulli.stan'
```

```r
fit <- mode$sample(
  data = data_complicated,
  seed = 1,
  chains = 2,
  iter_sampling = 2000,
  refresh = 0
)
```

```
## Running MCMC with 2 sequential chains...
##
## Chain 1 finished in 37.2 seconds.
## Chain 2 finished in 38.6 seconds.
##
## Both chains finished successfully.
## Mean chain execution time: 37.9 seconds.
## Total execution time: 76.1 seconds.
```

```r
fit
```

```
##      variable   mean median   sd  mad      q5    q95 rhat ess_bulk ess_tail
##  lp__         -97.35 -96.99 2.53 2.44 -102.06 -93.89 1.00     1234     1696
##  intercept      0.67   0.70 0.96 0.96   -0.93   2.18 1.00     4150     2887
##  slope_height   0.14   0.14 0.02 0.02    0.10   0.17 1.00     1727     2146
##  slope_weight  -0.05  -0.05 0.02 0.02   -0.09  -0.01 1.00     2579     2448
##  slope_pts      0.09   0.08 0.08 0.08   -0.04   0.21 1.00     2980     2312
```

```
##  slope_ast      0.10   0.10 0.24 0.24   -0.28   0.52 1.00     2564     2554
##  slope_draft   -0.53  -0.53 0.05 0.05   -0.62  -0.44 1.00     2480     2982
##  slope_age     -0.36  -0.35 0.11 0.11   -0.55  -0.18 1.00     2419     2639
##  slope_oreb    -0.04  -0.07 0.42 0.36   -0.69   0.65 1.00     1869     1955
##  center        -0.10  -0.10 0.17 0.14   -0.36   0.17 1.00     2147     1863
##
##  # showing 10 of 12 rows (change via 'max_rows' argument or 'cmdstanr_max_rows' option)
```

```r
fit$summary(variables = c("intercept", "slope_height","height_slope_positive"), "mean","sd", "rhat","es
```

```
## # A tibble: 3 x 6
##   variable               mean     sd   rhat ess_bulk ess_tail
##   <chr>                 <dbl>  <dbl>  <dbl>    <dbl>    <dbl>
## 1 intercept             0.671 0.962   1.000    4150.    2888.
## 2 slope_height          0.138 0.0216  1.00     1728.    2146.
## 3 height_slope_positive 1     0      NA          NA       NA
```

```r
quantile(fit$draws("slope_height"), c(0.025,0.975))
```

```
##       2.5%      97.5%
## 0.09841602 0.18240869
```

```r
mean(fit$draws("height_slope_positive"))
```

```
## [1] 1
```

```r
data_list <- list(
  N = nrow(rookie_data),
  height = rookie_data$player_height,
  y = rookie_data$draft_round
)
mod <- cmdstan_model("logistic_slope.stan")
fit <- mod$sample(
  data = data_list,
  seed = 1,
  chains = 2,
  iter_sampling = 2000,
  refresh = 0
)
fit$summary(variables = c("intercept", "slope","slope_positive"), "mean","sd", "rhat","ess_bulk", "ess_
cat("Exponentiated value of the slope coefficient is", exp(mean(fit$draws("slope"))),".")
quantile(fit$draws("slope"), c(0.025,0.975))
```

The effective sample size is around 500 for both variables.

We choose to use 2 chains so we could check whether the chain mixes well later. From the table, we can conclude that the mean of slope is 0.03, and the mean of `slope_positive`, an indicator variable is 0.98.

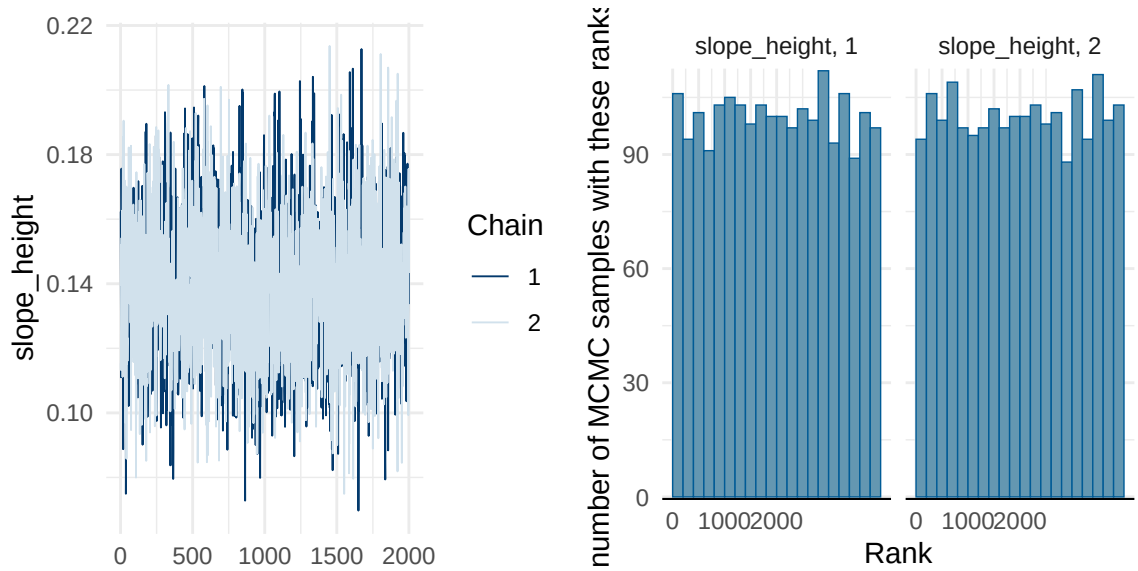
Since $E[1[slope > 0]|Y] = P(slope > 0|Y)$, this model effectively answered our question, as the probability of interest is substancially larger than 0.5. Thus, we conclude that height has a positive relationship with the chance of being drafted in the first round. Thus, being taller in height means higher probability of drafted earlier.

From the table, we can also conclude that a 1 cm increase in height is associated with a 2.6% increase in the odds of a player being drafted in the first round.

We would like to assess the goodness of fit using the following approach.

MCMC Diagnostics

```
trace <- mcmc_trace(fit$draws("slope_height")) + theme_minimal()
hist <- mcmc_rank_hist(fit$draws("slope_height")) +
  ylab("number of MCMC samples with these ranks") +
  theme_minimal()+
  scale_x_continuous(
  breaks = c(0, 500, 1000, 1500, 2000),
  labels = c("0", "", "1000", "", "2000"))
gridExtra::grid.arrange(trace,hist,ncol=2)
```



Here, we are using `slope_height` instead of `height_slope_positive` to check whether the chain mixes well. this is because that `slope` is a continuous variable , but `height_slope_positive` is binary. Thus, fitting the trace plots and histograms on this binary variable will most likely generate some vertical lines, which becomes harder to interpret. Since both variables are form the same model, if the plots for `slope_height` suggests fast mixing, then it means that the model for `height_slope_positive` mixes fast. From the trace plot, we can see that they two chains overlap a lot; from the histogram, we can see two approximately uniform histograms. Both plots suggests that our model is fast mixing.

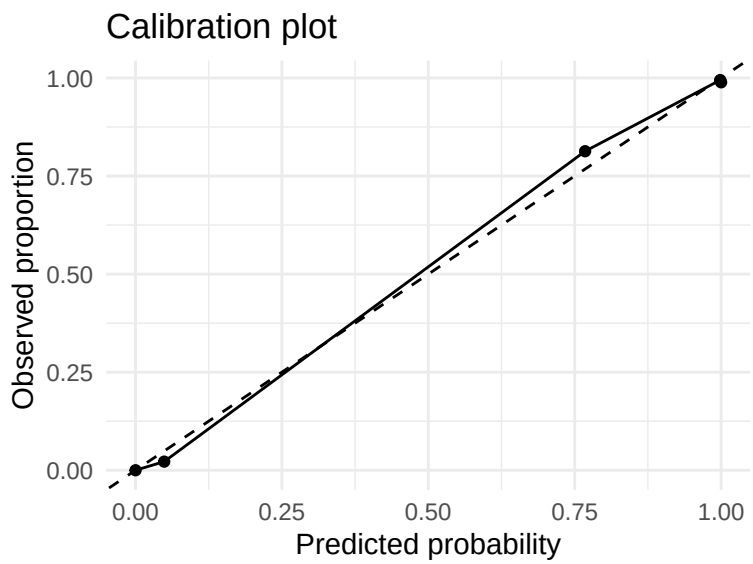
Goodness of Fit

```
draws_df <- as_draws_df(fit$draws(c("intercept", "slope_height", "slope_weight","slope_pts","slope_ast"

temporary_data$p_hat <- sapply(
  1:nrow(temporary_data),
  function(i) {
    mean(plogis(
```

```
      draws_df$intercept +
      draws_df$slope_height * temporary_data$player_height[i] +
      draws_df$slope_weight * temporary_data$player_weight[i] +
      draws_df$slope_pts * temporary_data$pts[i] +
      draws_df$slope_ast * temporary_data$ast[i] +
      draws_df$slope_draft * temporary_data$draft_number[i] +
      draws_df$slope_age * temporary_data$age[i] +
      draws_df$slope_oreb * temporary_data$oreb_pct[i]))})

temporary_data$bin <- ntile(temporary_data$p_hat, 5)
calibration_df <- temporary_data %>%
  group_by(bin) %>%
  summarise( mean_pred = mean(p_hat),
    obs_rate = mean(draft_round))
ggplot(calibration_df, aes(x = mean_pred, y = obs_rate)) +
  geom_point() + geom_line() +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  coord_cartesian() +
  labs(x = "Predicted probability", y = "Observed proportion",
    title = "Calibration plot") + theme_minimal()
```




```
draws_df <- as_draws_df(fit$draws(c("intercept", "slope")))
rookie_data$p_hat <- sapply(
  1:nrow(rookie_data),
  function(i) {
    mean(plogis(draws_df$intercept + draws_df$slope * rookie_data$player_height[i]))})
rookie_data$bin <- ntile(rookie_data$p_hat, 5)
calibration_df <- rookie_data %>%
  group_by(bin) %>%
  summarise( mean_pred = mean(p_hat),
    obs_rate = mean(draft_round))
ggplot(calibration_df, aes(x = mean_pred, y = obs_rate)) +
  geom_point() + geom_line() +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
```

```
  coord_cartesian(xlim = c(0.5, 0.65), ylim = c(0.5, 0.65)) +
  labs(x = "Predicted probability", y = "Observed proportion",
    title = "Calibration plot") + theme_minimal()
```

We can see that the majority of the probability center around 0.55 to 0.6, which suggests that although there is a positive relationship between height and draft round, the association is relatively weak.

Now, we would simulate the data and sere if the nominal coverage rate matches with the actual coverage rate. From, EDA, we see that the histogram of player height is approximately normal, centered roughly around 200cm, with a standard deviation around 10cm.

```
set.seed(405)
simulate_data <- function(N,intercept, slope){
  height <- rnorm(N,200,10)
  y <- rbinom(N, 1, plogis(intercept+slope*height))
  data <- list(N=N,height=height,y=y)
  fit <- mode$sample(
  data = data, chains = 2, iter_sampling = 1000,
  refresh = 0,show_messages = F)
  ci <- quantile(fit$draws("slope_height"), c(0.1,0.9))
return(ci)
}
iteration = 20; intercept = -1 ; slope = 0.01
covered <- numeric(iteration)
for (i in 1:iteration){
  ci <- simulate_data(200,intercept, slope)
  covered[i] <- (slope >= ci[1] & slope <= ci[2])
}
mean(covered)
```

```
simulate_data <- function(N,intercept,slope_height,slope_weight,slope_pts,slope_ast,slope_draft,slope_ag
  height <- rnorm(N,200,10)
  weight <- rnorm(N,100,10)
  draft_number <- rdunif(N,1,60)
  age <- rexp(N,1/80)
  pts <- rexp(N,1/10)
  ast <- rexp(N,1/5)
  oreb_pct <- runif(N,0,1)
  y <- rbinom(N, 1, plogis(intercept+
  slope_age*age +
  slope_height*height+
  slope_weight*weight+
  slope_draft*draft_number+
  slope_pts*pts+
  slope_ast*ast +
  slope_oreb*oreb_pct))

  data <- list(N=N,height=height,y=y,player_weight=weight,draft_number=draft_number,age=age,pts=pts,ast=
  fit <- mode$sample(
  data = data, chains = 2, iter_sampling = 1000,
  refresh = 0,show_messages = F)
  ci <- quantile(fit$draws("slope_height"), c(0.1,0.9))
return(ci)
```

```r
}

iteration = 1; intercept = -1 ; slope_height=0.1; slope_weight=-0.1; slope_pts=0.1; slope_ast=0.1; slope
covered <- numeric(iteration)
for (i in 1:iteration){
  ci <- simulate_data(200,intercept,slope_height,slope_weight,slope_pts,slope_ast,slope_draft,slope_age
  covered[i] <- (slope_height >= ci[1] & slope_height <= ci[2])
}
mean(covered)
```

```r
fit.variational = mode$variational(
  seed = 1,
  refresh =  0,
  output_dir = "stan_out",
  algorithm = "fullrank",
  output_samples = 1000,
  data = list(
    N = nrow(temporary_data),
    height = temporary_data$player_height,
    y = rookie_data$draft_round,
    draft_number = temporary_data$draft_number,
    age = temporary_data$age,
    player_weight = temporary_data$player_weight,
    pts = temporary_data$pts,
    ast = temporary_data$ast,
    oreb_pct = temporary_data$oreb_pct) , show_messages = F)
mean(fit.variational$draws()[, "slope_height"])
```

```
## [1] 0.2476149
```

```r
mean(fit.variational$draws()[, "height_slope_positive"])
```

```
## [1] 0.994
```

```r
quantile(fit.variational$draws()[, "slope_height"], c(0.025,0.975))
```

```
##       2.5%      97.5%
## 0.06112545 0.43433321
```

References

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